

# 1

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i = 1, \dots, n$$

## i

First find method of least squares, find the sum of residuals squared

$$Q = \sum_{i=1}^n \epsilon_i^2$$

$$Q = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

Take derivative with respect to  $\beta_0, \beta_1$  and set to 0

1.  $\frac{dQ}{d\beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) = 0$
2.  $\frac{dQ}{d\beta_1} = -2 \sum_{i=1}^n X_i (Y_i - \beta_0 - \beta_1 X_i) = 0$

Solve for  $\hat{\beta}_0$

$$\text{Using eq 1) } \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\rightarrow \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = \frac{0}{n} = 0$$

$$\rightarrow \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Plug this into eq 2

$$\sum_{i=1}^n X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\rightarrow \sum_{i=1}^n X_i (Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) - \hat{\beta}_1 X_i) = 0$$

$$\rightarrow \sum_{i=1}^n X_i Y_i - X_i \bar{Y} + X_i \hat{\beta}_1 \bar{X} - \hat{\beta}_1 X_i^2 = 0$$

$$\rightarrow \sum_{i=1}^n (X_i Y_i - X_i \bar{Y}) = \sum_{i=0}^n (\hat{\beta}_1 X_i^2 - X_i \hat{\beta}_1 \bar{X})$$

$$\rightarrow \sum_{i=1}^n (X_i Y_i - X_i \bar{Y}) = \hat{\beta}_1 \sum_{i=0}^n (X_i^2 - X_i \bar{X})$$

$$\rightarrow \frac{\sum_{i=1}^n (X_i Y_i - X_i \bar{Y})}{\sum_{i=0}^n (X_i^2 - X_i \bar{X})} = \hat{\beta}_1$$

ii

2

i

```
data = read.csv("data/slumpptest.csv", sep=",")
model = lm(CompressiveStrength ~ Water, data=data)
summary(model)
```

Call:

```
lm(formula = CompressiveStrength ~ Water, data = data)
```

Residuals:

| Min      | 1Q      | Median  | 3Q     | Max     |
|----------|---------|---------|--------|---------|
| -19.3590 | -5.4508 | -0.9858 | 4.6896 | 18.8254 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 55.48241 | 7.39784    | 7.500   | 2.54e-11 *** |
| Water       | -0.09861 | 0.03733    | -2.642  | 0.00956 **   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.618 on 101 degrees of freedom

Multiple R-squared: 0.06464, Adjusted R-squared: 0.05537

F-statistic: 6.979 on 1 and 101 DF, p-value: 0.009557

```
print(paste("Beta0 =", model$coefficients[1]))
```

```
[1] "Beta0 = 55.4824064174304"
```

```
print(paste("Beta1 =", model$coefficients[2]))
```

```
[1] "Beta1 = -0.0986112998623879"
```

**ii**

If there is no water content then the compressive strength of concrete is on average 55.482.

On average, the compressive strength of the concrete decreases by 0.09861 per unit increase of water content

**iii**

```
vcov(model)
```

|             | (Intercept) | Water        |
|-------------|-------------|--------------|
| (Intercept) | 54.7279850  | -0.274712638 |
| Water       | -0.2747126  | 0.001393292  |

**iv**

```
confint(model, level=0.95)
```

|             | 2.5 %      | 97.5 %     |
|-------------|------------|------------|
| (Intercept) | 40.8070888 | 70.1577241 |
| Water       | -0.1726577 | -0.0245649 |

Thus we are 95% confident that when the water content is 0 then the compressive strength of the concrete is between [40.807, 70.1577]

Thus we are 95% confident that the compressive strength of the concrete increases on average  $[-0.1726577, -0.0245649]$  per unit increase of water content.

**v**

$$H_0 : \beta_1 = -1, H_1 : \beta_1 < -1$$

$$\alpha = 0.1$$

Using the t-test:

$$t^* = \frac{\hat{\beta}_1 - c}{s(\hat{\beta}_1)} = \frac{-0.098611 + 1}{\sqrt{0.001393292}}$$

$$t^* = 24.1485568675$$

From summary,  $df = 101$ , and from above  $\alpha = 0.1$  so the critical  $t$  value is  $-1.29$  (left tailed)

Since this is left tailed, any  $t$  value greater than  $-1.29$  is rejection. Since  $24.1485568675 > -1.29$  we reject  $H_0$ .

Thus using the  $t$  test we can conclude that the engineer is wrong and the slope is not less than  $-1$ .

**vi**

```
predict(model, data.frame(Water=230))
```

```
      1
32.80181
```

**vii**

```
predict(model, data.frame(Water=230), interval="predict", level=0.95)
```

```
      fit      lwr      upr
1 32.80181 17.42295 48.18067
```